

Asynchronous timing of hypoxia and thermal stress induces diel vertical movement

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Abstract

1. Mobile organisms respond to multiple environmental variables and optimizing one habitat condition often comes at the expense of another. Trade-offs between temperature and dissolved oxygen are ubiquitous in aquatic ecosystems, and they are also dynamic, as the day-night cycle drives solar absorption and photosynthesis. Heat stress and hypoxia often occur asynchronously in the diel cycle, because temperature maximums occur in the evening and oxygen minimums occur around dawn. A key question is whether animal behavior can soften temperature-oxygen trade-offs by exploiting this asynchrony.
2. We tested how fish respond to diurnal changes in DO and temperature within vertically stratified thermal refuges on a river floodplain. First, we used empirical environmental data to simulate three behavioral strategies; a fish that minimizes temperature, a fish that maximizes DO, and a fish that optimizes both temperature and DO. Next, we held fish *in-situ* in enclosures and quantified their use of the thermal environment using temperature-logging tags coupled with arrays instrumented with temperature-oxygen loggers.
3. Our simulation showed that strategies that focused solely on either temperature-minimizing or oxygen-maximizing resulted in constant depth selection, whereas fish that optimized both temperature and oxygen exhibited diel vertical movement. The fish in our enclosure study exhibited diel vertical movement that matched the timing predicted by our simulation, thereby reducing the minimum oxygen levels they experienced. The extent of this movement was smaller than expected as fish endured relatively hypoxic conditions in exchange for colder temperatures, which reduce metabolic costs.
4. We show that diel vertical migration is an emergent property of temperature-oxygen optimization that exploits asynchrony between these stressors. This phenomenon caused

fish to experience more optimal combinations of temperature and DO than those found at single locations. Our conclusions are grounded in first principles and should therefore be generalizable across the diversity of aquatic systems in which analogous temperature-oxygen tradeoffs occur.

Keywords

Diel vertical migration, temperature-oxygen squeeze, trade-off, floodplain thermal refuge, spatiotemporal variation, stratified

Introduction

Environmental heterogeneity can decouple local habitat conditions from regional trends, creating asynchronous variation among habitat patches (Dobrowski, 2011; Fey et al., 2019). While spatial heterogeneity in resources underpins foundational concepts in ecology such as optimal foraging, competition, and niche theory, resource availability can vary both spatially and temporally. This added dimension increases complexity because animals must make decisions to optimize their well-being under conditions that are constantly changing. “Green wave surfing” or the behavior of following the leading edge of vegetation emergence, for example, has been observed among many terrestrial herbivores and allows animals to track more favorable conditions across the accessible landscape as they become seasonally available (Van der Graaf et al., 2006). While there is much interest in how animals behaviorally respond to spatio-temporal heterogeneity in single resources at large spatial scales (Armstrong et al., 2016; Abrahms et al., 2021), less is known about responses to multiple resources at fine spatial scales.

In aquatic ecosystems, both temperature and dissolved oxygen (DO) are critically important, because most aquatic organisms are poikilotherms that cannot regulate their body

temperature, and because oxygen is present at lower concentrations and is more variable in water than in air. Higher temperatures both reduce oxygen solubility in water and increase organismal oxygen demand, so we should expect that as temperatures exceed physiological optimums, behavioral responses should be shaped by both temperature and DO. Aquatic habitats ranging from pelagic seas to ponds can exhibit vertical stratification of temperature and oxygen (also referred to as vertical fronts; Logan et al., 2023). In tropical marine environments, suboptimally cool and hypoxic waters below the thermocline can limit fish to the shallower depths (Brill et al., 1999; Logan et al., 2023). In temperate lotic and lentic environments, warm surface temperatures can exceed physiological optima, and hypoxic conditions at depth can limit fish vertical distribution, creating a vertically compressed habitat known as a temperature-oxygen squeeze (Coutant, 1985). This phenomenon can also occur in fine scale patches of lentic habitat within a river floodplain. For example, thermally stratified cold-water alcoves often present a temperature-oxygen squeeze, effectively confining cold-adapted fish to $\leq 20\%$ of the water column (Barrett et al., 2024). In these habitats, cool water is introduced through upwelling hyporheic flows, which are often oxygen-depleted due to biological respiration and the absence of photosynthesis while subsurface. The vast majority of this work on temperature and oxygen stratification considers a temporally static trade-off. However, diel variation is a prominent feature in many systems as the day-night cycle modulates temperature and oxygen through solar absorption (Caissie, 2006) and ecosystem metabolism (Maasri et al., 2019).

To maximize fitness under diel variation in environmental variables, aquatic organisms often exhibit diel vertical migration (DVM). DVM is among the most widespread movement behaviors in aquatic systems, occurring in all sizes of water bodies and across multiple trophic levels. This behavior often takes advantage of asynchronous conditions. For example, fish

exploit asynchronous variation in predation risk and foraging opportunities through crepuscular ascents to surface waters (Scheuerell & Schindler, 2003). While DVM is a pervasive behavioral strategy that has been studied in response to diel variation in various trade-offs, there is virtually no knowledge of whether DVM or analogous habitat cycling can alleviate the trade-off between temperature and DO, which is critically relevant to understanding the effects of climate change.

To explore how behavioral responses can mediate the strength of dynamic temperature-oxygen trade-offs we: (1) monitored spatio-temporal heterogeneity in temperature and oxygen to characterize the diel and vertical heterogeneity, (2) simulated fish habitat selection under alternative tactics (minimize temperature, maximize oxygen, or optimize temperature and oxygen) to identify emergent behaviors, 3) recorded fish depth selection *in situ* by tagging fish with high-resolution temperature loggers and releasing them into large net pens to empirically test hypothesized behavioral responses, and 4) fit resource selection models to understand how fish selected for thermal conditions while controlling for changing availability of temperature and DO. We hypothesized that fish would exhibit diel vertical movements (DVM) to balance asynchronous variation in temperature and oxygen constraints. By combining simulations and empirical data, we tested alternative hypotheses about habitat selection strategies, providing critical insights into how fish navigate spatiotemporal variation in habitat conditions. As climate change intensifies, cold-water adapted fish will likely become increasingly reliant on thermal refuge habitat, and our study provides critical insights into how fish navigate spatiotemporal variation in habitat conditions

Methods

Study location

The Willamette River basin lies between the Cascade and Coastal Mountain ranges of Western Oregon. Historically, the Upper Willamette River (between Eugene and Salem) was a laterally dynamic and meandering gravel-bed river with side channels, islands, and gravel beds (Wallick et al., 2013; Flitcroft et al., 2023). While these features still exist, they have been significantly reduced since Euro-American settlement (Baker et al., 2004; Gregory et al., 2019). This portion of the river also contains numerous alcoves, i.e., floodplain features with only a downstream connection to the mainstem river. A subset of these waterbodies function as cold-water refuges, providing temperatures up to $\sim 10^{\circ}\text{C}$ cooler than the adjacent mainstem river due to a combination of hyporheic and surface water heat transfer processes (Arrigoni et al. 2008; Gombert et al., 2022). These cold-water habitats are seasonally exploited by cold-water fishes, particularly coastal cutthroat trout (*Oncorhynchus clarkii clarkii*). Cold-water alcoves are often characterized by pronounced vertical stratification in dissolved oxygen (DO; Zarnetske et al., 2011; Gombert et al., 2023) and diel oxygen cycling (Andersen et al., 2017). Surface water temperatures also fluctuate on a diel cycle in time-lagged synchrony with DO saturation (e.g. oxygen saturation is highest during midday, whereas surface water temperatures reach their maximum during early evening). The fluctuating conditions found in these alcoves likely cause the zone of optimal habitat to shift over the 24-h cycle. While temperature and oxygen fluctuate in rough synchrony (both increase during the day as water warms and photosynthetic activity generates oxygen and decrease at night due to radiative heat loss and respiration without photosynthesis), ideal conditions fluctuate asynchronously when considered through the lens of a cold-water fish, where cooler temperatures coupled with higher DO are ideal.

Spatiotemporal patterning of temperature and oxygen

113 In 2020, we collected data on temperature and DO within six known cold-water alcoves
114 on the Upper Willamette River to characterize spatiotemporal variation in DO and temperature,
115 and the resulting trade-offs to fish. During the summer, we instrumented each of these alcoves
116 with two separate temperature-DO arrays deployed laterally (i.e. along the length of the alcove)
117 near its mid-point and near the end of the alcove. Each array was comprised of three
118 temperature-DO loggers (Onset HOB0® U26-001 Dissolved Oxygen Data Logger; DO
119 accuracy to $\pm 0.2\text{mg}\cdot\text{L}^{-1}$, and temperature to $\pm 0.2^{\circ}\text{C}$) which were calibrated using 100% water
120 saturated air (Onset Computer Corporation, 2016). Loggers were set to record at 10-minute
121 intervals at the surface, mid-, and bottom of the water column. Arrays were deployed in each
122 alcove for a total of 3-4 days.

123 We then focused our study on a subset of these alcoves for more intensive monitoring to
124 explore both simulated and observed fish behavior. In 2021, we continuously monitored two of
125 these alcoves, Blue Ruin and Norwood, for DO and temperature over a ~3-week period during
126 peak-summer to obtain more spatially continuous temperature-DO data. We conducted this study
127 during peak-summer when thermal stress was most acute to examine how fish navigate
128 challenging environmental conditions that potentially represent a key vulnerability in their
129 habitat use. In Blue Ruin alcove, we deployed five vertical logger arrays laterally along the
130 deepest portions of the alcove to characterize spatiotemporal heterogeneity in conditions
131 (Supplemental Figure 1). Each array was instrumented with at least two temperature-DO loggers
132 positioned in the epilimnion (the warm, upper layer of a stratified water column) and
133 hypolimnion (the cold, bottom layer of a stratified water column). One of the arrays in Blue Ruin
134 (the site-4 array), as well as the array deployed in Norwood alcove, were associated with our fish
135 enclosure study, which is described in greater detail below. These arrays were more heavily

instrumented to obtain profile data at a finer resolution. Loggers on all arrays were set to record at 5-minute intervals. In 2021, we also recorded vertical profiles of temperature and DO during the early morning and evening to understand how these vertical gradients vary between low oxygen periods and peak oxygen periods. For simplicity, we hereafter will refer to the DO maximum as ‘day’ and DO minimum as ‘night’.

Simulating fish movements

We used the temperature and oxygen data from one of the arrays we deployed in Norwood alcove in 2021 to simulate how fish depth should change over a 24-hour cycle under three different behavioral strategies: a fish that selects habitat to maximize dissolved oxygen ($DO_{\max}$), a fish that selects habitat to minimize temperature ($T_{\min}$), and a fish that selects habitat to maximize a water quality index that includes both temperature and dissolved oxygen (TDO_{opt} , described below). We first selected a 24-hour period (July 29) that was representative of typical summertime conditions at this location. We then interpolated between loggers to create a more continuous prediction (i.e. finer resolution) of estimated temperature and DO set at every 0.1m and spanning the entire depth of the water column. Estimates were calculated using a modified slope equation (see Barrett et al., 2024).

To produce simulated responses to these habitat conditions, we first created habitat indices for the $T_{\min}$ and $DO_{\max}$ strategies. Our approach loosely follows methods detailed in (Sullivan et al., 2003). First, the highest and lowest DO concentrations recorded over the 24-hour period were ranked as most optimal (1) and least optimal (0), respectively. The lowest and highest recorded temperatures were similarly ranked (with the lowest temperature as most optimal (1)). We then used linear regression to calculate the slope and intercept values between

these two indices to determine the index values for the full range of temperature and oxygen recorded over this 24-hour timeframe.

We then combined the temperature and oxygen indices into a TDO_{opt} index that weighted these two variables equally using the equation:

$$TDO_{opt} = (Tmin_i * DOmax_i)^{1/2} \text{ (Sullivan et al. 2003)}$$

These three indices were then used to predict fish depth over the 24-hour period. For each strategy, we determined the highest index value at each time step and found its associated depth. We then plotted these depths, per strategy, over a 24-hour timeframe.

Next, we ran this simulation across the 7-day study window that we observed fish in enclosures, so that we could compare predicted to observed patterns of habitat use. Here we performed two separate analyses using the net pen array data from both cold-water alcoves. Once again we interpolated between logger depths to create a more fine-resolution prediction of temperature and DO across the full depth of the water column. Two of the temperature-DO loggers at the Blue Ruin site-4 net pen array (loggers at 1.55m and 0.5m depth) were fouled with algae during the study and frequently recorded excessively high and low DO concentrations (despite weekly cleaning of the sensors). The site-5 logger array (closest to the terminus of the alcove; Supplemental Figure 1), however, was within 10m of the site-4 net pen array and comparison of logger data between these two sites, as well as profile data from both locations, showed that recordings of temperature and oxygen were quite comparable throughout the vertical profile (depth and time-matched DO measurements were within $\leq 0.3 \text{ mg}\cdot\text{L}^{-1}$). We opted to remove the DO recordings from the two fouled loggers at the site-4 net pen array and to incorporate the DO data from the site-5 array recorded by the loggers at the closest aligning

depths (loggers at 1.35 and 0.4m depth respectively). This new dataset was used for the depth selection simulation, to quantify fish DO selection from our enclosure study, and for the habitat selection function, described in further detail below. We also used these data to determine the maximum daily temperature and minimum daily DO encountered by simulated fish under the three different behavioral strategies, which we plotted using boxplots.

Quantifying movement in situ

We sought to quantify the depths and associated temperature and oxygen conditions selected by fish in the wild. While pressure sensors on tags can estimate depth, they lack the accuracy needed for detecting sub-meter changes. Instead, we tagged fish with temperature loggers and inferred depth and oxygen from vertical profiles, following the approach of Barrett et al. (2024).

In the summer of 2022, we externally tagged 31 cutthroat trout with waterproofed Thermocron high-resolution iButtons (DS1922L-F5#, temperature accuracy $\pm 0.5^{\circ}\text{C}$). These iButtons were modified to reduce weight following methods outlined by Lovegrove (2009) and set to record temperature at 12-minute intervals to maximize data resolution while ensuring adequate battery life and data storage capacity. Each modified iButton was mounted to a dark green 2.5 cm Peterson disc using marine grade epoxy (JB MarineWeld™ syringe two-part adhesive). The Peterson discs had two small gauge holes drilled 3 mm apart and 3 mm in from the disk's outer margin, which were used to suture the mounted iButton to each fish through their anterior dorsal musculature. Tag weights ranged from 3.4 g to 5.3 g ($4.1 \text{ g} \pm 0.4$, mean $\pm$ SD) and did not exceed 3.8% of fish body mass (empirical studies have shown tag burdens $\leq 6.7\%$ do not negatively impact fish; Brown et al., 2010).

Cutthroat trout were captured via electrofishing on July 22 and 23, 2021. Anesthetization, handling, and recovery procedures followed Liedtke et al. (2012). Tags were attached through the anterior dorsal musculature of each fish approximately 5 mm ventral and 5 mm posterior of the first fin ray using a PDS® Plus 4-0 violet-dyed suture with a ½ circle 26-mm needle. After full recovery, the fish were transferred to net pens. We deployed three net pen enclosures, two in Blue Ruin alcove and one in Norwood alcove. Net pens were comprised of plastic mesh (1.27 cm x 1.27 cm) formed into a 1.68 m diameter cylinder that extended the full depth of the water column (~1.8-2.5 m). The bottom of the pen was weighted to the benthos and sewn closed with a panel of mesh. The surface of the pen extended ~0.45 m above the water's surface and was also capped with mesh. Floats were attached around the upper margins of the pen, creating a semi-rigid and secured cylindrical pen that extended the full depth of the water column.

We placed 11 tagged fish in the Norwood alcove enclosure and 20 in the Blue Ruin enclosures (10-fish per enclosure). Stocking densities ranged from 0.8 to 0.87 kg/m³ to reduce fish stress and maintain welfare. Research on stocking densities in caged Atlantic salmon (*Salmo salar*) found that fish welfare (determined by blood glucose, cortisol, and body condition) began to decrease at densities above 22 kg/m³ (Turnbull et al., 2008). In total, we collected data from 13 fish in Blue Ruin alcove and 8 fish in Norwood alcove. Nine fish escaped the net pens during the study, and two of the iButtons recovered from Blue Ruin were corrupted.

To characterize temporal shifts in the vertical distribution of temperature and DO, we deployed a vertical array of DO and temperature loggers outside and within 1-meter of the net pen. A temperature-DO profile was first taken using a handheld YSI (ProSolo Digital handheld sonde, Yellow Springs, OH, USA; calibrated daily) to determine depth of the thermocline at 12:00 noon. Temperature-DO loggers (MiniDOT® Logger, dissolved oxygen accuracy to ±

225 0.1mg•L⁻¹ and temperature to $\pm 0.1^{\circ}\text{C}$, or Onset HOBO® U26-001 Dissolved Oxygen Data
226 Logger dissolved oxygen accuracy to $\pm 0.2\text{mg}\cdot\text{L}^{-1}$, and temperature to $\pm 0.2^{\circ}\text{C}$) were set to read
227 at 25-cm below the surface, at the thermocline, and at the hypolimnion. Additional temperature
228 loggers (HOBO Water Temperature Pro v2 Data Logger, temperature accuracy to $\pm 0.2^{\circ}\text{C}$) were
229 affixed to the line between the bottom and mid-dissolved oxygen logger, between the mid and
230 surface dissolved oxygen logger, and at the water's surface. Net pens were cleared of any debris
231 or buildup twice weekly but were otherwise left undisturbed. Logger membranes were also
232 wiped clean during this maintenance. At the end of the study, fish were removed from the pen,
233 temperature loggers were removed, and fish were released back into the alcove.

234 *Analysis*

235 *Quantifying diel changes in temperature and oxygen trade-offs*

236 We combined all array data from the summers of 2020 and 2021. Based on time series
237 plots of the data, we identified 6:00-8:00 as the best overlapping 2-hour window of minimum
238 DO, and 18:00-20:00 as the best overlapping 2-hour window of maximum DO, across all
239 alcoves. To characterize trade-offs in the relationship between temperature and DO, and how
240 they varied among alcoves and between the windows of oxygen minima and maxima, we used
241 linear mixed effect models following Barrett et al. (2024). We fit separate models for day (the
242 oxygen maxima) and night (the oxygen minima). To account for random variation from sample
243 site, we included alcove location as a random intercept. Year was not included as a random
244 intercept because it was not a significant source of variation in the model. We also fit separate
245 linear models to the data collected at our two focal alcoves, Blue Ruin and Norwood. All models
246 were fit using the lme4 package in R (Bates et al., 2014).

Quantifying movement patterns

Fish with externally mounted iButtons often exhibited temperatures that fell in between the temperatures recorded at the net pen associated logger arrays. We used linear interpolation to create a continuous estimation of temperature and DO (at 0.2m increments) for both net pen associated arrays. We then matched fish iButton temperatures to the closest bounding profile temperatures for each time step and used linear interpolation to gain resolution in our fish-depth estimates (Barrett et al., 2024). We used this same approach to estimate the DO concentration for each calculated fish depth. We used these data to determine the maximum daily temperature and minimum daily DO encountered by tagged fish in each cold-water alcove habitat, which were plotted using boxplots. Our analysis was limited to data collected ≥ 5 -days post fish capture to ensure cortisol levels and swimming behavior had returned to normal (Jepsen et al., 2005).

Habitat selection function

The cyclic and covarying nature of temperature and DO levels within a 24-hour period means that simply characterizing the conditions that fish use provides an incomplete understanding of how changing conditions are mediated by behavioral adjustments such as diel vertical migration. For example, observing that fish used areas in the water column with a temperature of 16°C tells us nothing about the preference of fish for certain temperatures until we compare the temperature profiles chosen to what was available, i.e. selection. We therefore sought to quantify what temperatures fish selected for by quantifying the temperatures fish used at a given time of day in proportion to the available temperatures within the water column that a fish could choose from at the same moment in time. To do this, we used resource selection functions in a ‘used-available’ design as is routinely done in habitat selection studies in terrestrial systems (Manly et al., 2012). Specifically, we used conditional logistic regression (sometimes

referred to as matched case-control logistic regression or discrete choice models). For each iButton reading, we compared the temperature and DO at these ‘used’ locations to temperature and DO values in 10 random locations (sampled from a uniform distribution spanning the entire depth of the water column) that were ‘available’ for a fish to use at that moment in time, but were not selected. Therefore, the values of our response variable were 1 if the datapoint corresponded to a used location, and 0 if corresponded to an available location. We used these temperature-DO profile sets to calculate the relative probability of selection, given the range of temperature and DO available. We restricted our analysis to the two-hour periods of lowest and highest DO in Blue Ruin and Norwood alcoves and comprised 1-week of data (July 30 to August 6, commencing 5-days after fish were released into net pens). The result was four model sets, one for each data subset. Temperature and DO were scaled to have mean = 0 and sd = 1 and were the sole predictors in the model given our objective was to assess how fish selected for temperature while controlling for both the available temperature and DO conditions. Because we anticipated that the response of fish to changing conditions might be nonlinear, we did not want to constrain the relationship with a priori assumptions about its functional form. Therefore, we formulated our conditional logistic regression as a generalized additive model. We fit models using the `cox.ph` function (using REML to estimate smoothing parameters, (Wood, 2004)) in the R package `mgcv` (Wood 2015) because conditional logistic regression has the same likelihood as a cox proportional hazards model (Manly et al., 2012).

We then created prediction data frames where temperature was allowed to vary across the full range of values recorded, while DO was held constant at $4\text{mg}\cdot\text{L}^{-1}$, a cutoff representing the value used by the State of Oregon to indicate stress in “cool water aquatic life”; Oregon Department of Environmental Quality, 2007). We next predicted the marginal probability of

selection for temperature using the predict() function in R, with type = "link" and se.fit = TRUE. Finally, we plotted predictions of the relative probability of selection for temperature and their 95% confidence intervals across the span of temperature-DO combinations selected by tagged fish.

Results

Vertical and diel variation in temperature and DO

Vertical profiles of temperature in both Blue Ruin and Norwood alcoves showed pronounced stratification, with higher contrasts between layers during daytime (Figures 1 & 2). At night, water temperatures remained cool and within a relatively narrow range. During the day, bottom temperatures at both alcoves remained low (~13-13.5°C), while surface water temperatures reached levels similar to or slightly warmer than the mainstem river (~22 – 25°C). The vertical range of DO differed between sites. At night in Blue Ruin, DO ranged from ~2.5 mg•L⁻¹ at the bottom to 3.5 mg•L⁻¹ at the surface. At night in Norwood, DO was more vertically variable, ranging from 0.79 mg•L⁻¹ at the bottom to 7.0 mg•L⁻¹ at the surface. During the day DO spanned from 5.0 – 9.4 mg•L⁻¹ in Blue Ruin and 8.8 – 11.1 mg•L⁻¹ in Norwood. Vertical temperature profiles taken along the Blue Ruin alcove thalweg were consistently similar, but the DO profiles varied more across the alcove. For instance, at night, maximum epilimnetic DO ranged from 3.7 to 6.7 mg•L⁻¹, while minimum hypolimnetic DO dropped as low as 0.3 mg•L⁻¹ in certain areas.

Both Blue Ruin and Norwood cold-water alcoves displayed distinct diel cycling of DO and temperature (Figure 2, Supplemental Table 1, Supplemental Figure 2 & 3). In both alcoves, a DO minima and maxima occurred during the early morning ('night') and early evening ('day'),

respectively. However, the 2-hour periods of high and low DO occurred at slightly shifted windows of time (~5:00-7:00 & ~17:00-19:00 for Blue Ruin, ~6:00-8:00 & ~18:00-20:00 for Norwood), likely due to differing alcove orientation and solar exposure.

We observed a clear trade-off between cool temperature and high DO, as illustrated by a positive linear relationship between the two variables, expressed across six alcoves during both the morning and evening (Figure 3, Supplemental Table 2). However, the shape of the trade-off differed between day and night. In all alcoves there was a contracted range of both temperature and DO, meaning fish had reduced potential to trade-off cool temperature for higher DO. At five of the six alcoves, the slope was also more positive at night, meaning that a given level of cooling was accompanied by a larger reduction in DO (i.e., the trade-off was stronger). Notably, we found the opposite result in the Blue Ruin alcove, where the water column, at night, was more homogeneously low in temperature and DO. This meant that fish had limited potential to trade off temperature for oxygen during the period of DO minimum.

Diel patterns of fish depth predicted by alternative behavioral responses

Our simulation yielded three distinct patterns of fish depth use across the diel cycle (Figure 4a). A fish that maximized dissolved oxygen ($DO_{\max}$) would always hold near the water's surface, while a fish that minimized temperature ($T_{\min}$) would always hold near the bottom. In contrast to these stationary behaviors, a fish that considered both temperature and oxygen (TDO_{opt}), exhibited diel vertical movements; shifting towards the surface during the late night and early morning when DO at depth became limiting, and then moving deeper during the day when surface temperatures were warmer and deeper habitats became more oxygenated. Thus, our simple simulation demonstrated how DVM is an emergent property in systems where

temperature and DO present spatial trade-offs that vary over the diel cycle (i.e., due to solar absorption, photosynthesis, and respiration).

Fine scale diel vertical movements in captive fish

Cutthroat trout monitored *in situ* did not hold at constant depth, but instead exhibited diel vertical movement analogous to that predicted in our simulation (Figure 4b). All individuals expressed DVM, shifting 0.2-1.3m shallower during the night. A smoothed fit of fish depth versus time-of-day, pooled across all individuals, yielded a distinct vertical movement pattern (Figure 4c). Fish shifted deeper in the water column during the morning (~6:00) and held at depth until early evening (~18:00). Interestingly, these vertical movements did not span the entire water column and were constrained roughly to the hypo- and metalimnion. By avoiding the warmer epilimnion, tagged fish encountered maximum daily temperatures and minimum daily DO that fell between those selected by TDO_{opt} and T_{min} simulated fish (Figure 5).

Shifting selection probability as temperature and oxygen fluctuate

Our resource selection models supported disparate responses to selection for the thermal environment between the windows of maximum and minimum DO. After controlling for changing availability of both temperature and DO, tagged fish in both cold-water alcoves selected for warmer temperatures during the night and cooler temperatures during the day, relative to what was available during those periods (Figure 6, Supplemental Table 3). These results provide additional evidence that fish exhibited DVM to exploit asynchrony between temperature maximums and DO minimums. There was high correlation between the temperature and DO covariates (≥ 0.8), however this level of correlation was unavoidable as these two variables fluctuate in rough synchrony and are the cause of the tradeoffs (Figure 3). We

mitigated this correlation structure by focusing our inference on model predictions using marginal plots (Figure 6).

Discussion

There is growing interest in how organisms track spatiotemporal variation in resource availability (Abrams & Ginzburg, 2000), yet the overwhelming majority of this research focuses on individual resources that change in isolation. A much more complex situation arises when two critical resources vary spatiotemporally and asynchronously such that constant behavioral modifications are required to stay within conditions suitable for maintaining basic physiological functions. Therefore, as climate change continues to produce stressful conditions for thermally-intolerant species, it will be increasingly important to understand how organisms respond not only to changing temperatures, but also the suite of factors that covary with thermal conditions and may be equally important for maintaining fitness. Here we characterized diel variation in trade-offs between temperature and DO in a heat-sensitive species across six floodplain thermal refuges on a seasonally warm river. During the day, when temperature increased toward its maximum and photosynthesis increased DO concentrations, the temperature-DO trade-off was flatter and there was more oxygen overall (Figure 3). During the night, temperatures were cooler but respiration continually decreased DO, generating a steeper temperature-DO trade-off with less oxygen overall. Thus, the threats of hypoxia and heat stress were asynchronous and suggest that fish had to constantly adjust to the changing conditions.

By simulating behavior under alternative strategies, we showed that fish in systems characterized by asynchronous risk in heat stress and hypoxia should respond to the emergent temperature-oxygen constraints by utilizing DVM (Figure 4a). We empirically corroborated this hypothesis in wild cutthroat trout held *in situ*. Fish exhibited fine-scale DVM, moving shallower

during the day and deeper at night (Figure 4b, 4c). Statistical modeling confirmed that fish in these thermal refuges were avoiding cooler temperatures at night, when hypoxia was most severe, but were selecting cooler temperatures during the day, when DO was higher (Figure 6). Thus, diel variation, and adaptive fish behavior (DVM), soften the constraints of DO on thermal refuge use, also known as the ‘temperature-oxygen squeeze’. Temperature-DO trade-offs are common in freshwater ecosystems and our results are supported by first principles, so they are likely generalizable to other aquatic ecosystems.

Our simulation of fish habitat selection found that fish held at constant depth if considering temperature or DO in isolation but switched to diel vertical movement when considering temperature and DO together (Figure 4a). Tagged fish held *in situ* exhibited analogous DVM with similar timing, but at a smaller amplitude than in our simulation (Figure 4b, 4c). We did not expect our simulation to accurately predict fish depths in the wild, because the TDO_{opt} strategy was not based on empirical physiological data; i.e., we do not know the actual costs of ephemeral exposure to differing levels of hypoxia as there can be high intraspecific variation in hypoxia tolerance (Faust et al., 2004; Metcalfe et al., 2016). However, the fish in our study appear to be relatively tolerant of temporary exposure to low DO. Though they avoided the lowest DO levels in the water column, they could have experienced higher DO minimums by moving farther above the thermocline. For example, the EPA criteria for absolute minimum DO levels for coldwater fish is $4 \text{ mg}\cdot\text{L}^{-1}$ (USEPA, 1987), the cutthroat in our study could have stayed above this level by selecting $\sim 2\text{--}3^\circ \text{C}$ warmer water, but they did not. Matthews and Berg (1997) found similar patterns where rainbow trout aggregated near groundwater seeps, which provided cooler water with lower DO in stratified pools. Despite the

availability of warmer surface waters with higher DO, the trout preferred cooler, oxygen-poor refugia.

We were surprised that fish did not select for temperatures warmer than what we observed in favor of higher DO, because recent physiological work on our focal species found that temperatures up to 19°C did not decrease aerobic scope (Anlauf-Dunn et al., 2022). For example, the average maximum temperature selected by fish in the Blue Ruin net pens was ~16.5°C even though average maximum available temperatures recorded at the Blue Ruin net pen logger array reached 22.7°C and would have provided higher DO. While short-term thermal stress does not appear to explain the observed behavior in our fish, we believe cumulative effects on energetics do. While sampling fish in cold-water alcoves as part of a separate study, we observed high densities of fish with small ration sizes comprised of low energy density prey items (unpublished data). When fish are starving, cooler temperatures reduce rates of energy loss (by reducing metabolic costs) and increase the duration over which fish can survive without foraging (Fry, 1947). Thus, when fish balance temperature-DO trade-offs on thermal refuges, where negative energy budgets are typical (Hahlbeck et al., 2022), incurring some level of hypoxia may reduce the risk of starving to death during periods of thermal stress. Rainbow trout can display metabolic depression in response to diel hypoxia cycling (Williams et al., 2019), and repeated exposure to thermal cycling can alter fish metabolic phenotype and increase energy sequestration in the liver (Callaghan et al., 2016). These physiological tactics may further reduce energy loss, but more work on the energetics of refuge use is needed before this can be established as a general phenomenon.

Tagged fish displayed high individual and intra-individual variation in their vertical movements, however, the central tendency was consistent with a vertically cycling movement

pattern constrained to ~20% of the water column, ranging between the metalimnion and upper hypolimnion (Figure 4c). This behavioral variation could reflect variation in genetics (Alonzo, 2015), individual thermal tolerance (Anlauf-Dunn et al., 2022) or energy reserves (Moran et al., 2021). Fish ration size affects the optimal temperature for growth and could influence habitat selection by fish (Mac, 1985; Sogard & Olla, 1996). However, previous diet data collected in these alcoves suggest that fish in alcoves barely feed and do not meet energetic demands (Barrett unpublished data), so captive fish (which likely had reduced access to food or may cease feeding due to captivity) likely did not have substantially lower energy intake compared to fish in the wild, and thus should have similar thermal habitat selection in the context of ration.

In lotic systems, spatiotemporal variation in temperature and DO is driven by both external drivers like atmospheric conditions (Caissie, 2006) and internal structures like hydrologic connectivity (Poole & Berman, 2001; Palmer & Ruhi, 2019), topography (Isaak & Hubert, 2001), and vegetation (Beschta, 1997). In cold-water alcoves, the patterns of DO and temperature can vary both within and among habitats. While diel variation and vertical stratification were common to all sites, there was variation in both the maximum and minimum values of temperature and oxygen. For example, 15 of our 22 logger arrays recorded average minimum hypolimnetic DO of $<2 \text{ mg}\cdot\text{L}^{-1}$ at night, whereas during the day, average maximum hypolimnetic DO levels at those same sites ranged from $3.9 \text{ mg}\cdot\text{L}^{-1}$ to $12.4 \text{ mg}\cdot\text{L}^{-1}$. Oxygen concentrations $< 3 \text{ mg}\cdot\text{L}^{-1}$ are often considered hypoxic, however some cold-adapted fish have shown decreased swimming capacity at DO concentrations of $\leq 7 \text{ mg}\cdot\text{L}^{-1}$ (Evans, 2007). This variability in DO indicates that fish may encounter severely hypoxic conditions at night, while daytime conditions offer a wider range of oxygen levels. However, the magnitude and nature of

these trade-offs also differed substantially between sites. Indeed, there was notable variation in the severity of these trade-offs among our two primary study sites, Norwood and Blue Ruin.

In both Norwood and Blue Ruin alcove, there was a high probability of fish selecting the coolest temperatures (at the greatest depths) during the day, when DO was high (Figure 6). At night, during the DO minima, the probability of selection switched to warmer water (towards the surface) where DO was also higher. Site-specific variation in temperature and DO may explain the varied strength (slope) of selection probability. For example, the diel amplitude of hypolimnetic DO was much higher in the Norwood alcove, while epilimnetic DO stayed relatively high across that timeframe (Supplemental Table 1, Supplemental Figure 2). However, surface temperatures in Norwood remained high even at night. These surface temperatures may have been suboptimally warm to fish, thereby reducing the strength of selection towards the warmer and more oxygenated surface temperatures (Figure 6b). In Blue Ruin, temperatures in the epilimnion were much lower at night (and DO remained adequate; Supplemental Table 1), which may explain the greater selection strength towards warmer (and shallower) temperatures when compared to Norwood (Figure 6b, 6d). Thus, not only did our fish display DVM, but they also exhibited flexibility in their movement patterns based on site-specific combinations of temperature and oxygen.

Here, we show that DO availability in floodplain thermal refuge habitats constrains vertical habitat use by a cold-water fish, adding to a growing body of work demonstrating similar habitat compression due to temperature-oxygen constraints in lakes and reservoirs (Coutant, 1985; Swales, 2006; McLaren et al., 2023), lentic habitats on a river floodplain (Barrett et al., 2024), and in some marine habitats (Deutsch et al., 2023). We encourage future research to address how diel variation in temperature and DO, and behavior such as DVM, could potentially

mitigate these synergistic constraints in other species and systems. Our novel continuous monitoring approach to could be applied in other studies to demonstrate whether diel variability in habitat conditions can cause trade-offs that drive cyclical habitat use in other systems besides floodplain thermal refuge habitats. Our work used both simulation and empirical data to address key questions about the temporal dynamics of vertical habitat compression and emphasizes the importance of considering how fine-scale diel fluctuation may impact the viability of thermal refuges under threat from climate change.

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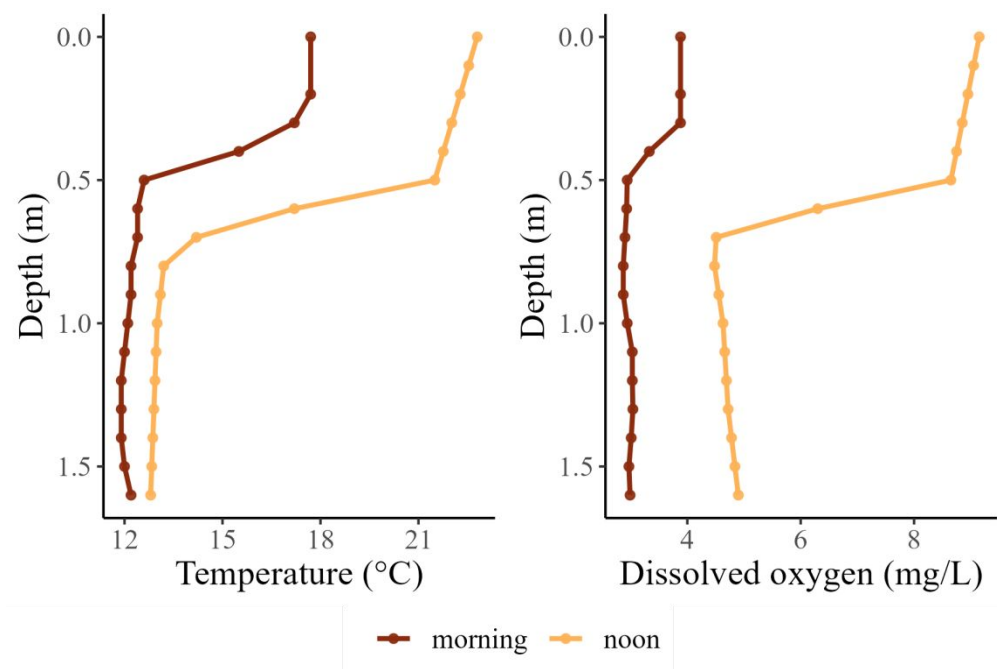


Figure 1 Diel contrasts in temperature and dissolved oxygen profiles for a floodplain alcove that provides thermal refuge for cold-water fishes of the Willamette River. Profiles were recorded at dawn (6:00, near DO-minima), and late afternoon (15:00, near the DO-maxima). Data are shown for the Blue Ruin alcove.

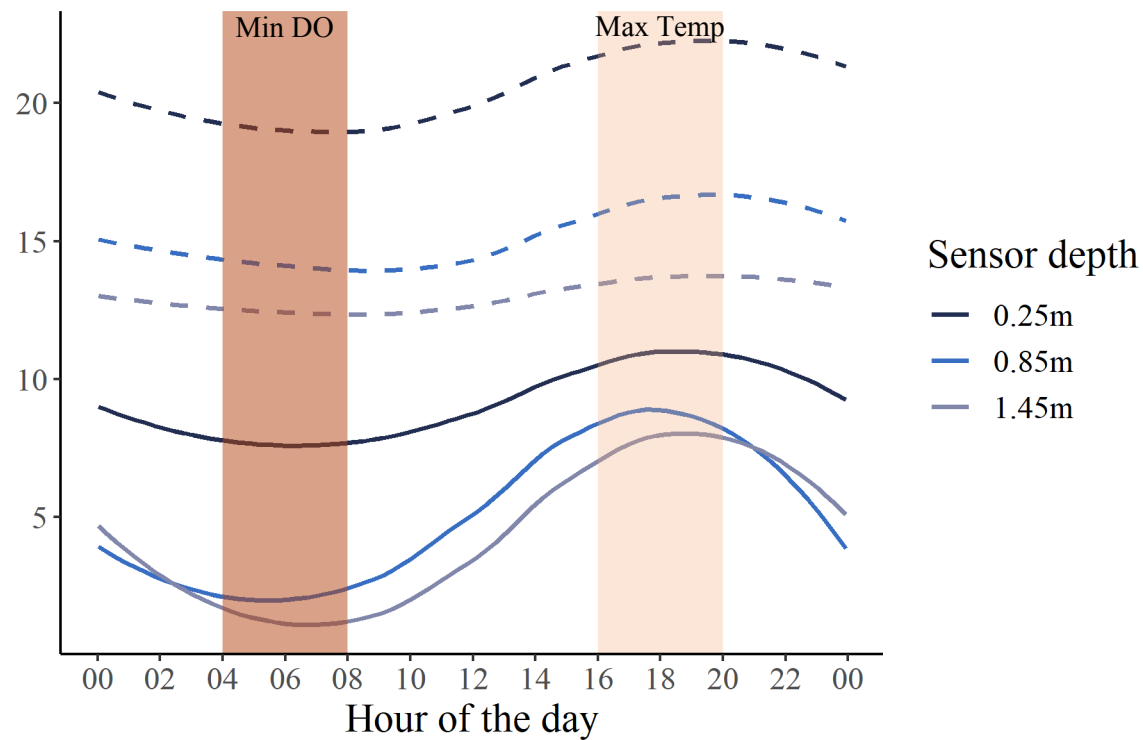


Figure 2 The diel cycle of water temperature and dissolved oxygen at three depths of a stratified floodplain alcove. Shaded bars highlight the asynchrony between periods of peak hypoxia (dark pink) and peak thermal stress (light pink). Data shown here are for July 27th 2021 in Norwood alcove, and have a 5-minute temporal resolution. Dashed lines represent temperature (°C), and solid lines represent dissolved oxygen (mg/L).

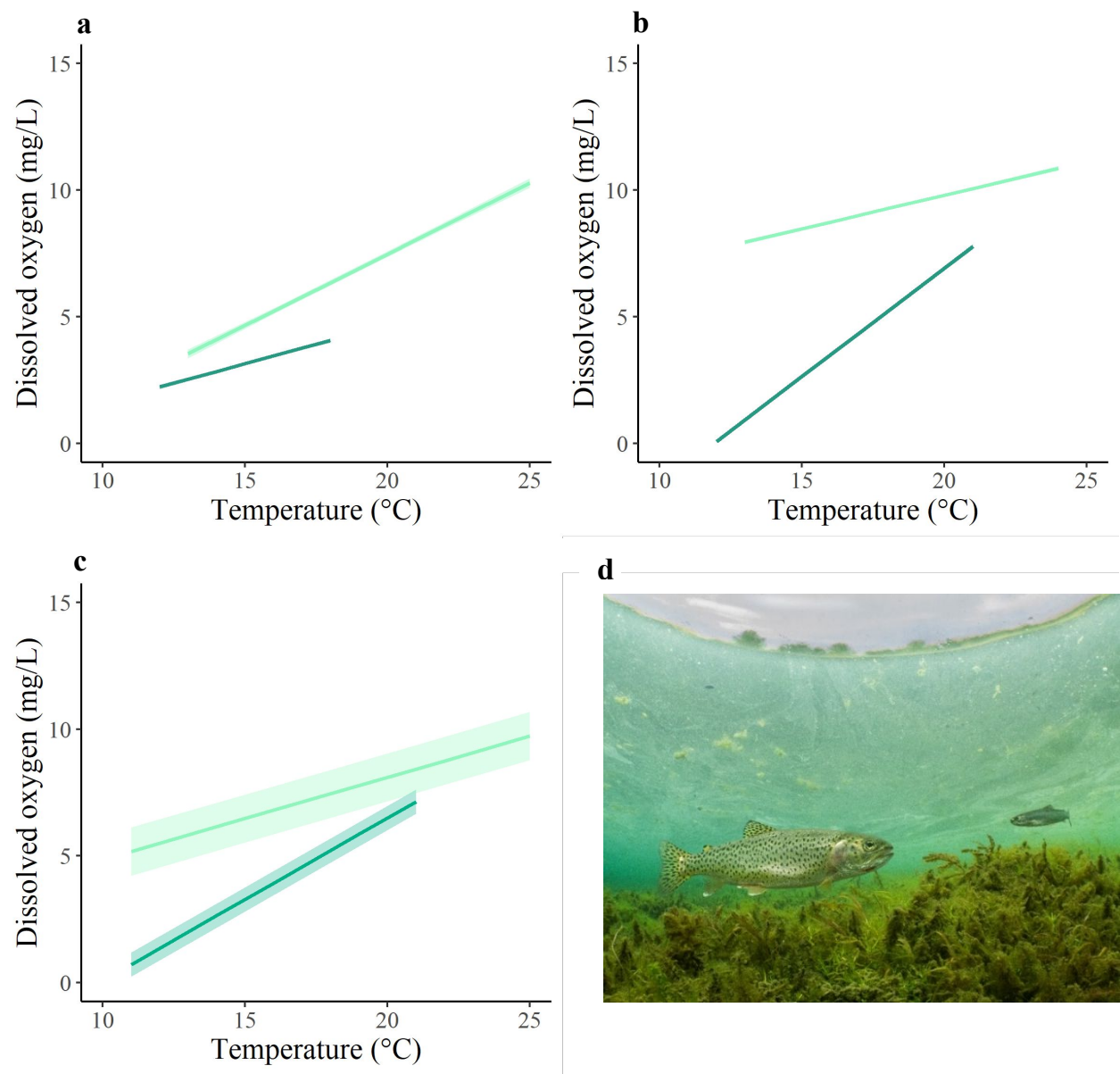


Figure 3 Results from linear mixed models evaluating the relationship between temperature and DO comparing the 2-hour window that approximates the DO-minima (dark green) to the 2-hour window that approximates the DO-maxima (light green) in **a**) Blue Ruin cold-water alcove, **b**) Norwood cold-water alcove, and **c**) all six cold-water alcoves. All linear fits are plotted with their 95% confidence interval envelopes. **d**) Photo of a coastal cutthroat trout in Blue Ruin cold-water alcove (photo credit: Dave Herasimtschuk, Freshwaters Illustrated).

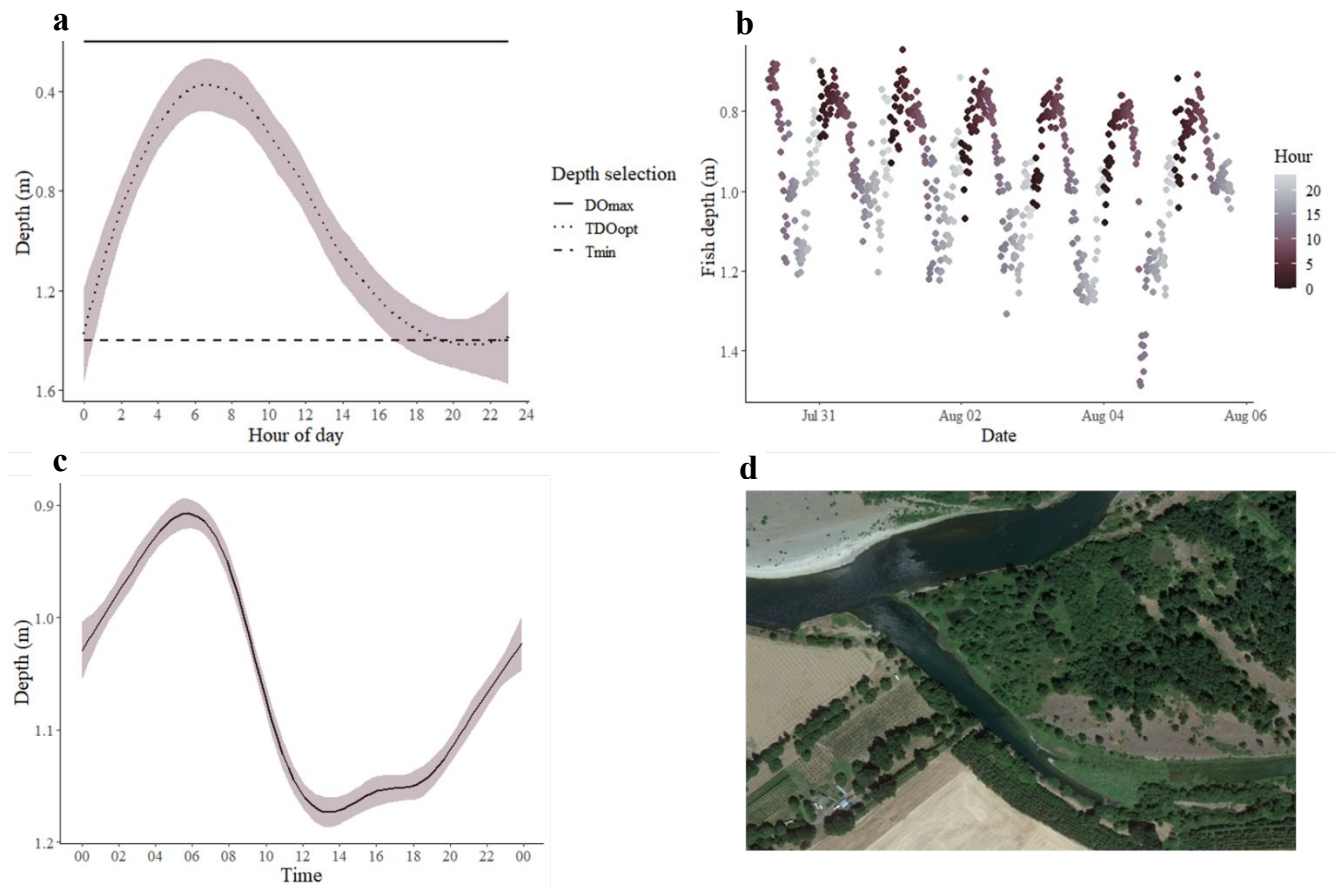
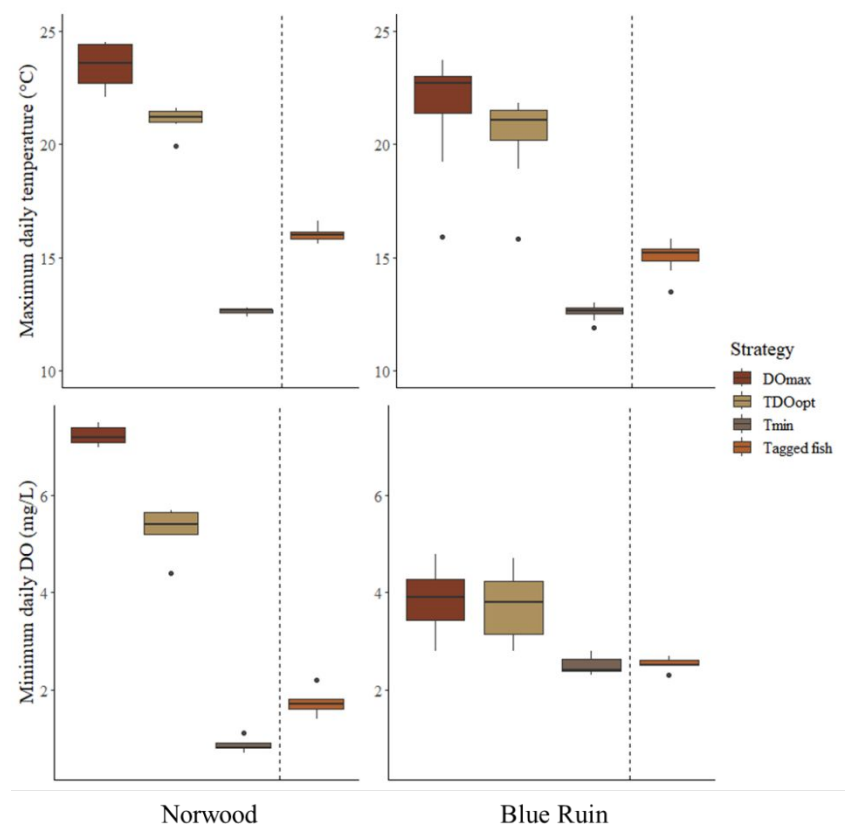


Figure 4a) Simulated predictions of fish depths throughout the diel cycle, based on three different strategies; maximizing dissolved oxygen (DO_{max}), minimizing water temperature (T_{min}), or considering both temperature and DO concurrently (TDO_{opt}). **4b)** Example of observed diel vertical migration for a tagged individual cutthroat trout held *in situ*. See Methods for how depth was inferred from tag temperature). **4c)** Smoothed plot of fish depth over 24-hour timeframe, pooled across all individuals in study. **4d)** Satellite image of Blue Ruin alcove (placeholder).



5) Maximum daily temperature and minimum daily DO encountered by simulated fish under three different strategies (n = 7 days), compared to corresponding average maximum and average minimum values for tagged fish held in enclosures (n = 19 individuals).

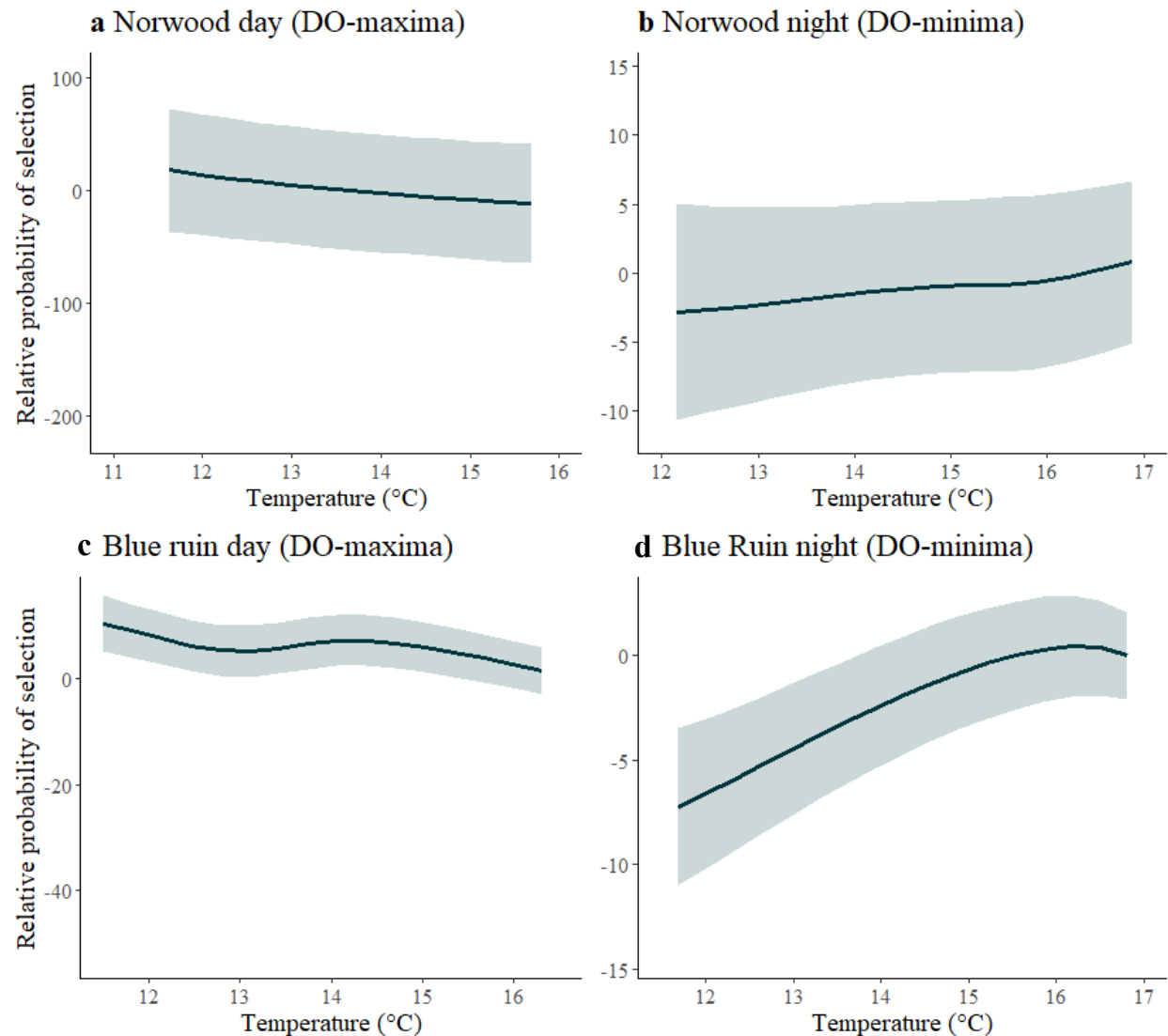
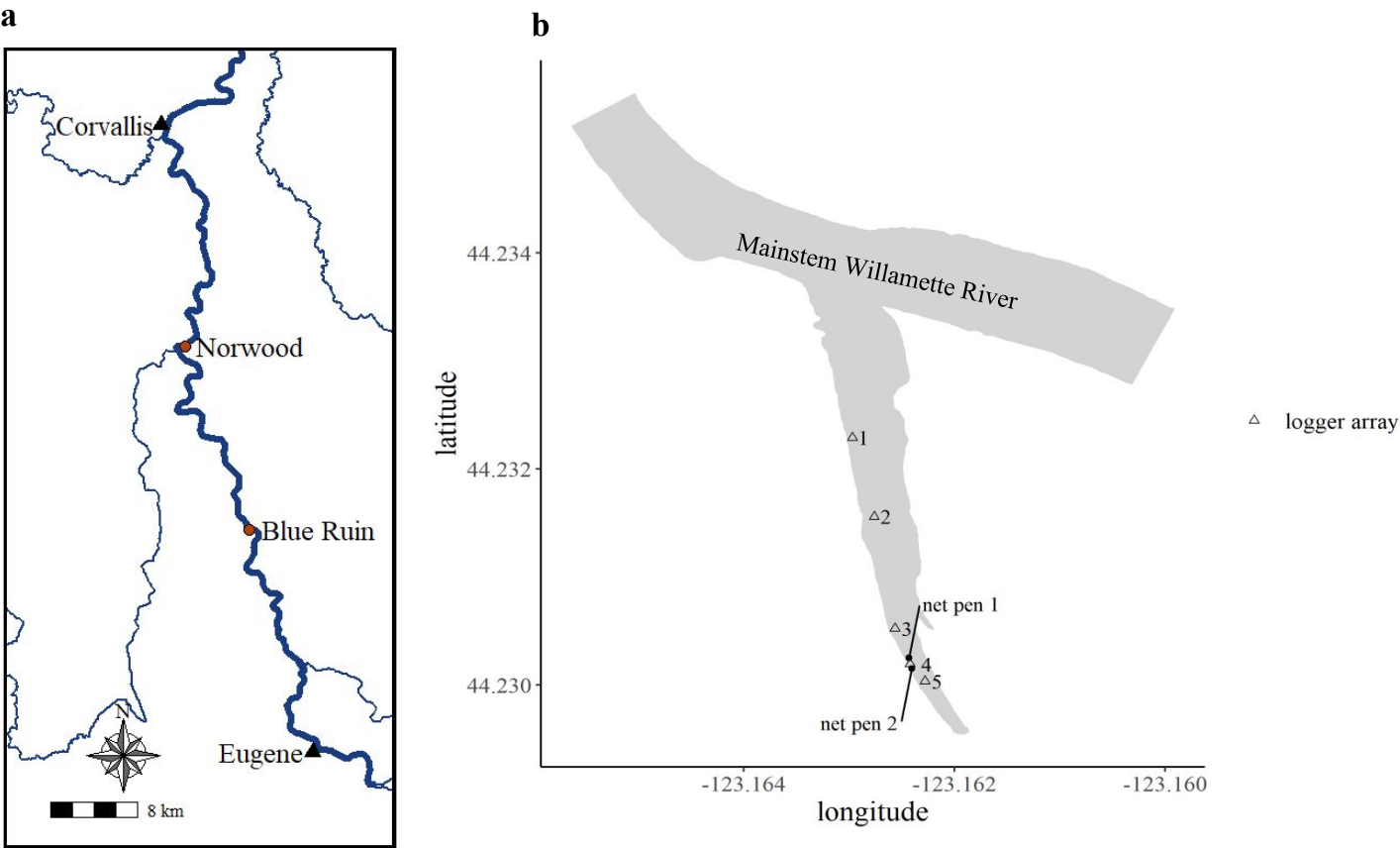
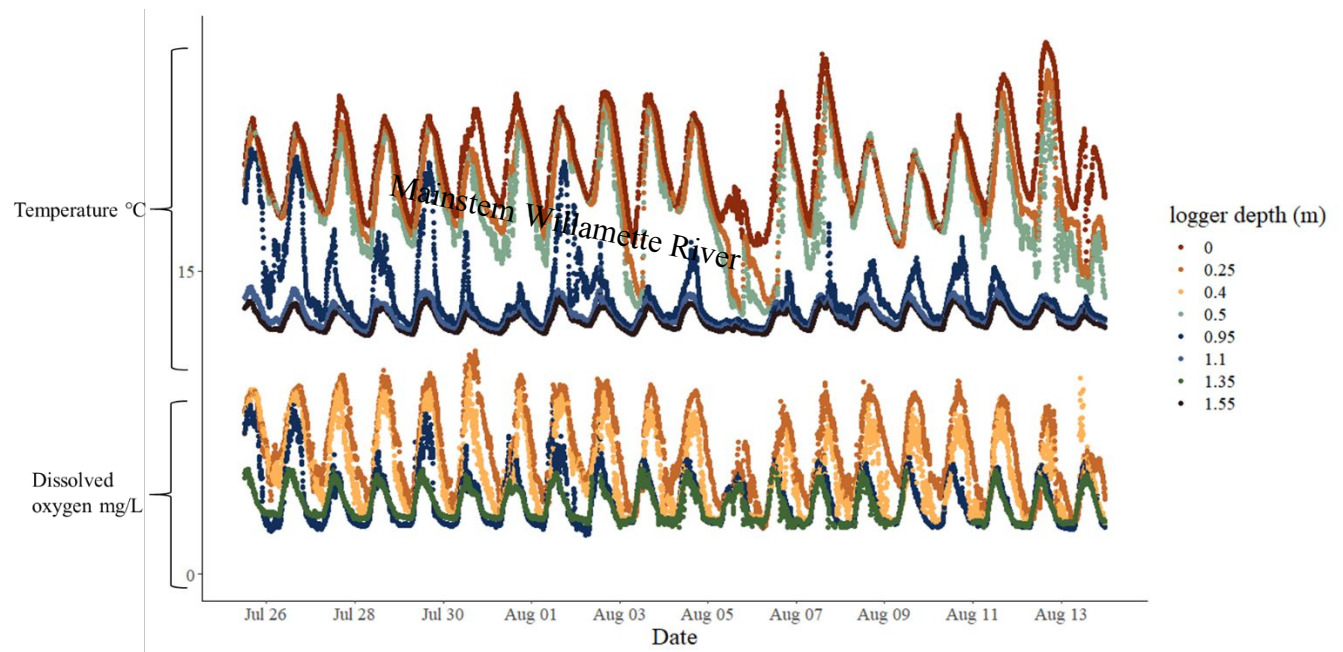


Figure 6 Predicted temperature selection for tagged coastal cutthroat trout held in enclosures during the 2-hour windows of DO-maxima and DO-minima. In all plots, DO was held constant at $4\text{mg}\cdot\text{L}^{-1}$ (EPA threshold for cold-water fish), and the x-axis was constrained to the range of temperatures selected by fish during the 2-hour window.

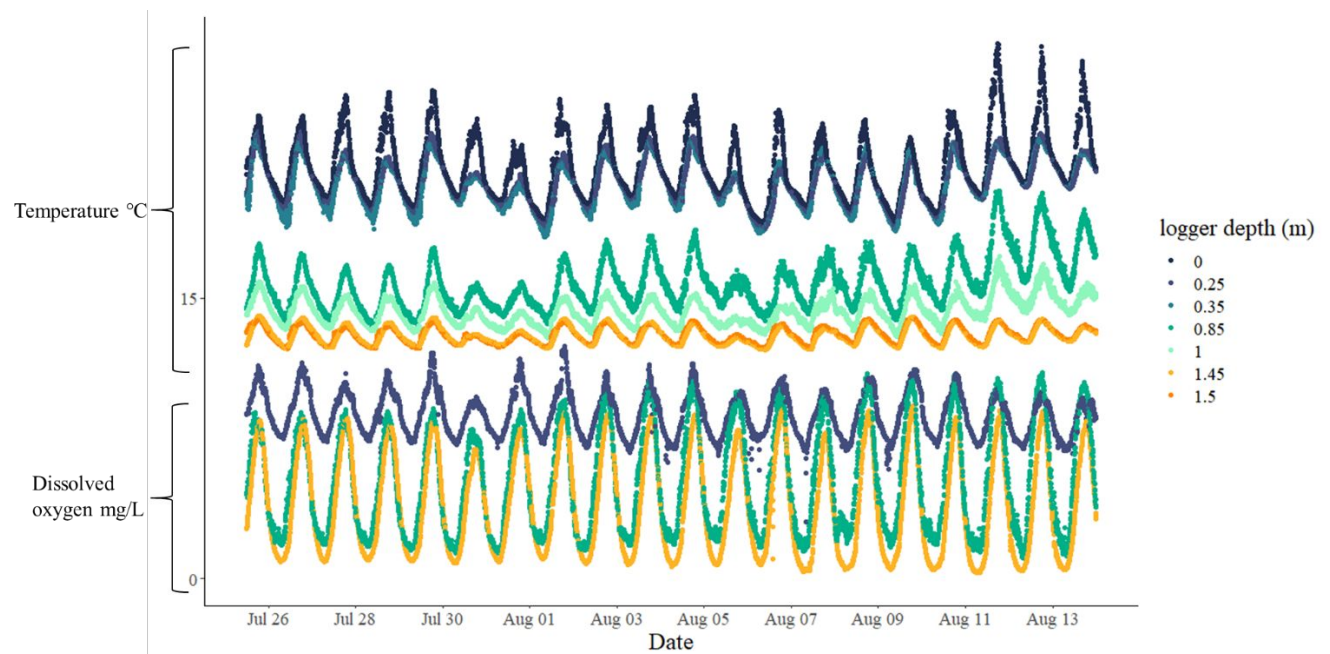
Supplemental Figures



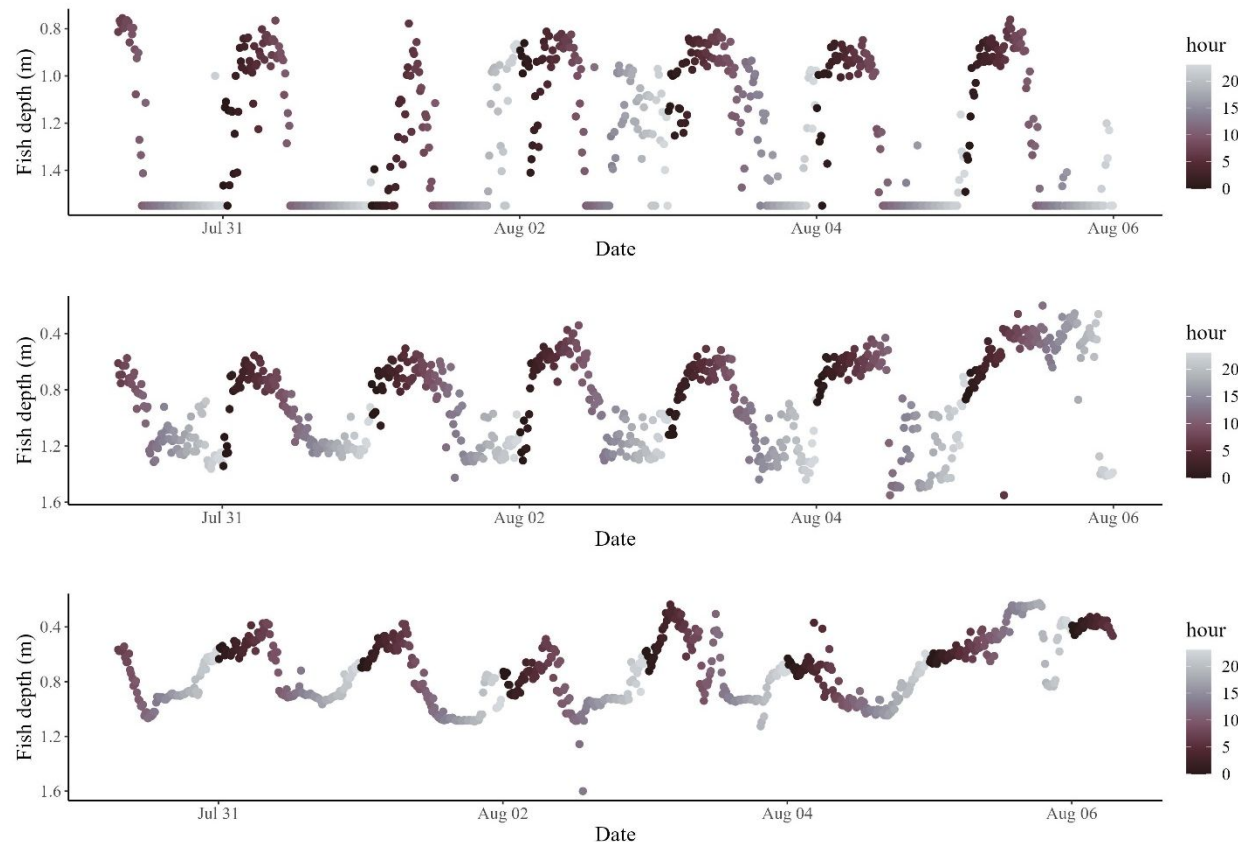
Supplemental Figure 1a Map of surveyed cold-water alcoves along the Upper Willamette River. Red points indicate cold-water alcove locations and triangles mark the location of major urban areas. **b** Figure of Blue Ruin cold-water alcove with logger array (numbered triangles) and net pen (black points) locations.



Supplemental Figure 2 Temperature and dissolved oxygen recorded at the Blue Ruin cold-water alcove site-4 net pen logger array.



Supplemental Figure 3 Temperature and dissolved oxygen recorded at the Norwood cold-water alcove net pen logger array.



Supplemental Figure 4 Depth plots of 3-individual tagged fish from the iButton netpen study, fish display individual and intra-individual level variation in movement, however all display fine scale diel vertical movement patterns.

Supplemental Table 1 Average minimum and maximum temperature and dissolved oxygen (DO) recorded at the hypolimnion and epilimnion in both Blue Ruin and Norwood alcoves at their net pen associated logger arrays.

Norwood 07-31 to 08-07	Temp (°C)	DO (mg/L)
Average minimum hypolimnion	12.5	1.2
Average maximum hypolimnion	13.6	8.4
Average minimum epilimnetic	20.0	6.9
Average maximum epilimnetic	25.0	11.3

Blue Ruin 07-31 to 08-07	Temp (°C)	DO (mg/L)
Average minimum hypolimnion	12.1	2.5
Average maximum hypolimnion	13.4	4.8
Average minimum epilimnetic	16.2	3.4
Average maximum epilimnetic	22.7	9.2

Supplemental Table 2 Results from linear models (Norwood, Blue Ruin) and linear mixed models (all CWA) fit to data collected during the approximate 2-hour DO-maxima (6:00-8:00), and 2-hour DO-minima (18:00-20:00).

		β	SE	p-value
Norwood	high DO	0.264	0.007	<0.001
	low DO	0.856	0.008	<0.001
Blue Ruin	high DO	0.563	0.013	<0.001
	low DO	0.306	0.011	<0.001
All CWA	high DO	0.325	0.009	<0.001
	low DO	0.642	0.008	<0.001

Supplemental Table 3 Summary of Generalized Additive Model (GAM) results fit using the smooth term $s(\text{standardized.do}, \text{standardized.temp})$. The table presents the effective degrees of freedom (EDF), reference degrees of freedom (RDF), Chi-squared statistic, and p-value.

Model	EDF	RDF	Chi-squared	p-value
Norwood day (DO-maxima)	13.4	14.82	153.4	< 2e-16
Norwood night (DO-minima)	14.39	16.53	350.5	< 2e-16
Blue Ruin day (DO-maxima)	14.28	15.76	472.8	< 2e-16
Blue Ruin night (DO-minima)	19.03	21.78	317.7	< 2e-16